Process Control Engineering & Loop Rationalization Manual

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Department:	Process Engineering & Instrumentation	**Site Location:**	Jamnagar Refinery Complex (JRC)

1. Architecture Overview

Defines control loop hierarchy from distributed control systems (DCS) to Safety Instrumented Systems (SIS). Establishes split-range, cascade, and override control strategies.

2. Regulatory Loop Performance KPIs

- Loop in Automatic Mode: Minimum 98.0% across all units.
- Control Valve Travel Reversals: Less than 15 reversals per hour to prevent gland packing wear.
- Output Saturation: Control valves must operate within 15% to 85% normal stroke span.

3. Alarm Rationalization Standard (EEMUA 191)

Average alarm rate per console operator must not exceed 1 alarm per 10 minutes during steady-state processing.